

UNIVERSITÀ DEGLI STUDI DI TORINO

Corso di Laurea Magistrale in Matematica



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Teoria delle categorie

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Introduzione

Testo

Capitolo 1

Teoria

1.1 The pullback functor

In this section we will define the pullback functor and its left adjoint, the shrink functor.

Given a category $\mathcal{A}$ with pullbacks and a morphism $p : E \rightarrow A$ in $\mathcal{A}$, one can consider adjunction

$$(\mathcal{A} \downarrow B) \quad \begin{array}{c} \xleftarrow{p^*} \\ \perp \\ \xrightarrow{p!} \end{array} \quad (\mathcal{A} \downarrow E)$$

defined as follows:

- $(\mathcal{A} \downarrow B)$ is the category of objects over B in $\mathcal{A}$, we denote an object by $(A, \alpha : A \rightarrow B)$;
- $(\mathcal{A} \downarrow E)$ is the category of objects over E in $\mathcal{A}$, we denote an object by $(D, \delta : D \rightarrow E)$;
- $p^* : (\mathcal{A} \downarrow E) \rightarrow (\mathcal{A} \downarrow B)$ is the pullback functor

$$\begin{array}{ccc}
 \begin{array}{c} A \\ \downarrow \alpha \\ B \end{array} & \xrightarrow{p^*} & \begin{array}{ccc} E \times_B A & \xrightarrow{\text{proj}_2} & A \\ \downarrow \text{proj}_1 & & \downarrow \alpha \\ E & \xrightarrow{p} & B \end{array}
 \end{array}$$

$$\begin{array}{ccc}
 \begin{array}{ccc} A & \xrightarrow{f} & A' \\ \downarrow \alpha & \swarrow \alpha' & \\ B & & \end{array} & \xrightarrow{p^*} & \begin{array}{ccccc} & & \exists! \tilde{f} & & \\ & \text{---} \text{dashed} \text{---} & & \text{---} \text{dashed} \text{---} & \\ E \times_B A & \xrightarrow{\text{proj}_2} & A & \xrightarrow{f} & A' & \xleftarrow{\text{proj}'_2} & E \times_B A' \\ & \searrow \alpha & \swarrow \alpha' & & \\ & B & & & \\ \text{proj}_1 \swarrow & p \uparrow & \searrow \text{proj}'_1 & & \\ & E & & & \end{array}
 \end{array}$$

- $p_! : (\mathcal{A} \downarrow B) \rightarrow (\mathcal{A} \downarrow E)$ is the shrink functor

$$\begin{array}{ccc}
 \begin{array}{c} D \\ \downarrow \delta \\ E \end{array} & \xrightarrow{p_!} & \begin{array}{ccc} D & & \\ \delta \downarrow & \searrow p\delta & \\ E & \xrightarrow{p} & B \end{array} \\
 \\
 \begin{array}{ccc} D & \xrightarrow{d} & D' \\ \delta \downarrow & \searrow \delta' & \\ E & & \end{array} & \xrightarrow{p_!} & \begin{array}{ccc} D & \xrightarrow{d} & D' \\ & \searrow p\delta & \downarrow p\delta' \\ E & \xrightarrow{p} & B \end{array}
 \end{array}$$

- the unit η^p

$$\begin{aligned}
 \eta : \text{Id}_{(\mathcal{A} \downarrow E)} &\rightarrow p^* p_! \\
 \eta_{(D, \delta)} = \langle \delta, 1_D \rangle : (D, \delta) &\rightarrow (E \times_B D, \text{proj}_1)
 \end{aligned}$$

$$\begin{array}{ccccc}
 D & & & & D \\
 & \searrow \exists! \langle \delta, 1_D \rangle & & \searrow 1_D & \\
 & E \times_B D & \xrightarrow{\quad} & D & \\
 \delta \downarrow & \downarrow & & \downarrow p\delta & \\
 E & \xrightarrow{p} & & B &
 \end{array}$$

- the counit ϵ^p

$$\begin{aligned}
 \epsilon : p_! p^* &\rightarrow \text{Id}_{(\mathcal{A} \downarrow B)} \\
 \epsilon_{(A, \alpha)} = \text{proj}_2 : (E \times_B A, p(\text{proj}_1)) &\rightarrow (A, \alpha)
 \end{aligned}$$

$$\begin{array}{ccc}
 E \times_B A & \xrightarrow{\text{proj}_2} & A \\
 \downarrow \text{proj}_1 & & \downarrow \alpha \\
 E & \xrightarrow{p} & B
 \end{array}$$

Let us show that this is actually an adjunction. First we prove that the unit is a natural transformation. Let $g : (D, \delta) \rightarrow ((D', \delta'))$ be a morphism in $(\mathcal{A} \downarrow E)$, we want to prove that the following diagram commutes:

$$\begin{array}{ccc}
 D & \xrightarrow{\eta_{(D, \delta)}} & E \times_B D \\
 \downarrow g & & \downarrow p^* p_!(g) = \tilde{g} \\
 D' & \xrightarrow{\eta_{(D', \delta')}} & E \times_B D'
 \end{array}$$

In order to do that we need to expand the action of the functor $p^* p_!$ on the morphism g . We have:

$$\begin{array}{ccc}
 \begin{array}{ccc} D & \xrightarrow{g} & D' \\ \delta \downarrow & \searrow \delta' & \\ E & & \end{array} & \xrightarrow{p_!} & \begin{array}{ccc} D & \xrightarrow{g} & D' \\ & \searrow p\delta & \downarrow p\delta' \\ & & B \end{array}
 \end{array}$$

$$\begin{array}{c}
 \xrightarrow{p^*} \quad \begin{array}{c}
 \begin{array}{ccccc}
 & & \exists! \tilde{g} & & \\
 & \swarrow & \text{---} & \searrow & \\
 E \times_B D & \xrightarrow{\text{proj}_2} & D & \xrightarrow{g} & D' & \xleftarrow{\text{proj}'_2} & E \times_B D' \\
 & \nwarrow \eta_{(D,\delta)} & \searrow p\delta & \swarrow p\delta' & \nwarrow \eta_{(D',\delta')} & & \\
 & & B & & & & \\
 & \nwarrow \text{proj}_1 & \uparrow p & \swarrow \text{proj}'_1 & & & \\
 & & E & & & &
 \end{array}
 \end{array}
 \end{array}$$

Considering the diagram

$$\begin{array}{ccccc}
 D & \xrightarrow{\eta_{(D,\delta)}} & E \times_B D & & \\
 \downarrow g & \searrow \text{---} & \downarrow p^* p_!(g) = \tilde{g} & & \\
 D' & \xrightarrow{\eta_{(D',\delta')}} & E \times_B D' & \xrightarrow{\text{proj}'_2} & D' \\
 & & \downarrow \text{proj}'_1 & & \downarrow p\delta' \\
 & & E & \xrightarrow{p} & B
 \end{array}$$

we can prove the naturality in the left diagram by the universal property of the pullback in the right. Since $(\text{proj}_1, \text{proj}_2)$ are jointly monic, by the following compositions

- $(\text{proj}'_2)(\tilde{g})(\eta_{(D,\delta)}) = (g)(\text{proj}_2)(\eta_{(D,\delta)}) = (g)(1_D) = g$
- $(\text{proj}'_2)(\eta_{(D',\delta')})(g) = (1_{D'})(g) = g$
- $(\text{proj}'_1)(\tilde{g})(\eta_{(D,\delta)}) = (\text{proj}_1)(\eta_{(D,\delta)}) = \delta$
- $(\text{proj}'_1)(\eta_{(D',\delta')})(g) = (\delta')(f) = \delta$

we conclude that $(\tilde{g})(\eta_{(D,\delta)}) = (\eta_{(D',\delta')})(g)$. So the naturality condition of η is satisfied.

Now we want to prove the counit is a natural trasfromation. Consider $f : A \rightarrow A'$ a morphism in $(\mathcal{A} \downarrow B)$, we want to prove that the following diagram commutes:

$$\begin{array}{ccc}
 E \times_B A & \xrightarrow{\epsilon_{(A,\alpha)}} & a \\
 p_! p^*(f) = \tilde{f} \downarrow & & \downarrow f \\
 E \times_B A' & \xrightarrow{\epsilon_{(A',\alpha')}} & A'
 \end{array}$$

As done similarly before we need to the action of the functor $p_!p^*$ on the morphism f , we have:

$$\begin{array}{ccc}
 \begin{array}{c} A \xrightarrow{f} A' \\ \downarrow \alpha \quad \swarrow \alpha' \\ B \end{array} & \xrightarrow{p^*} & \begin{array}{c} \begin{array}{ccccc} & & \exists! \tilde{f} & & \\ & \curvearrowright & & \curvearrowright & \\ E \times_B A & \xrightarrow{\text{proj}_2} & A & \xrightarrow{f} & A' & \xleftarrow{\text{proj}_2'} & E \times_B A' \\ & \searrow \alpha & \downarrow & \swarrow \alpha' & \\ & & B & & \\ & \nearrow \text{proj}_1 & \uparrow p & \nwarrow \text{proj}_1' & \\ & & E & & \end{array} \end{array} \\
 & & \xrightarrow{p_!} & \begin{array}{c} \begin{array}{ccccc} & & \exists! \tilde{f} & & \\ & \curvearrowright & & \curvearrowright & \\ E \times_B A & \xrightarrow{\text{proj}_2} & A & \xrightarrow{f} & A' & \xleftarrow{\text{proj}_2'} & E \times_B A' \\ & \searrow \alpha & \downarrow & \swarrow \alpha' & \\ & & B & & \\ & \nearrow p(\text{proj}_1) & & \nwarrow p(\text{proj}_1') & \\ & & E & & \end{array} \end{array}
 \end{array}$$

Since $\epsilon_{(A,\alpha)}$ is the $\text{proj}_{2,E \times_B A}$, the composition $(f)(\epsilon_{(A,\alpha)}) = (\epsilon_{(A',\alpha')})(\tilde{f})$ holds by the definition of $\tilde{f}$. So also the naturality of ϵ is satisfied.

Now we can check that the unit and counit fulfil the first triangle identities. Let (A, α) be an object in $(\mathcal{A} \downarrow B)$, we have:

$$((p^*\epsilon) \circ (\eta p^*))_{(A,\alpha)} = (p^*\epsilon)_{(A,\alpha)}(\eta p^*)_{(A,\alpha)} = p^*(\text{proj}_{2,E \times_B A})(\eta_{(E \times_B A, \text{proj}_1)}) = \circledast$$

Considering first how p^* acts on the morphism $\text{proj}_{1,E \times_B A}$

$$\begin{array}{c}
 \begin{array}{ccc} E \times_B A & \xrightarrow{\text{proj}_{2,E \times_B A}} & A \\ \searrow \alpha(\text{proj}_{2,E \times_B A}) & & \swarrow \alpha \\ & B & \end{array} \\
 \\
 \begin{array}{ccccc} & & \text{proj}_{2,\tilde{E \times_B A}} & & \\ & \curvearrowright & & \curvearrowright & \\ E \times_B (E \times_B A) & \xrightarrow{\text{proj}_{2,E \times_B (E \times_B A)}} & E \times_B A & \xrightarrow{\text{proj}_{2,E \times_B A}} & A & \xleftarrow{\text{proj}_{2,E \times_B A}} & E \times_B A \\ & \searrow (\alpha)(\text{proj}_{2,E \times_B A}) & \downarrow & \swarrow \alpha & \\ & & B & & \\ & \nearrow \text{proj}_{1,E \times_B (E \times_B A)} & \uparrow p & \nwarrow \text{proj}_{1,E \times_B A} & \\ & & E & & \end{array}
 \end{array}$$

$$\circledast = (\tilde{\text{proj}}_2)(\eta_{(E \times_B A, \text{proj}_1)})$$

and then considering the definition of η , we obtain

$$\begin{array}{ccccc}
 & & 1_{E \times_B A} & & \\
 & \nearrow & & \searrow & \\
 E \times_B A & & & & E \times_B A \\
 & \searrow \exists! \langle \text{proj}_1, E \times_B A, 1_{E \times_B A} \rangle & & \nearrow & \\
 & E \times_B (E \times_B A) & \longrightarrow & E \times_B A & \\
 & \downarrow & & \downarrow p(\text{proj}_1, E \times_B A) & \\
 \text{proj}_1, E \times_B A \nearrow & E & \xrightarrow{p} & B & \nwarrow
 \end{array}$$

$$\circledast = (\tilde{\text{proj}}_2)(\langle \text{proj}_1, E \times_B A, 1_{E \times_B A} \rangle)$$

In order to conclude we need to check some compositions so that, by the universal property of the pullback $E \times_B A$ in the right of the following diagram, we obtain the identity morphism.

$$\begin{array}{ccccccc}
 & & 1_{E \times_B A} & & \tilde{\text{proj}}_2 & & \\
 & \nearrow & & \searrow & & \nearrow & \\
 E \times_B A & \xrightarrow{\langle \text{proj}_1, E \times_B A, 1_{E \times_B A} \rangle} & E \times_B (E \times_B A) & \longrightarrow & E \times_B A & \xrightarrow{\text{proj}_2, E \times_B A} & A \longleftarrow E \times_B A \\
 & \searrow & \downarrow & & \downarrow (\alpha)(\text{proj}_2, E \times_B A) & \swarrow \alpha & \\
 & & & & B & & \\
 & \nearrow & & \searrow & \uparrow p & \nearrow \text{proj}_1, E \times_B A & \\
 & & \text{proj}_1, E \times_B (E \times_B A) & & E & & \\
 & \nwarrow & & \nearrow & & \nwarrow & \\
 & & \text{proj}_1, E \times_B A & & & &
 \end{array}$$

$$\circledast = (\tilde{\text{proj}}_2)(\langle \text{proj}_1, E \times_B A, 1_{E \times_B A} \rangle) = \text{Id}_{(E \times_B A, \text{proj}_1)}$$

We want to check that composition of the projections of $E \times_B A$ with $\circledast$ is the same of composing those projection with $\text{Id}_{(E \times_B A, \text{proj}_1)}$. Following the first composition

$$\begin{aligned}
 (\text{proj}_2, E \times_B A)(\tilde{\text{proj}}_2)(\langle \text{proj}_1, 1_{E \times_B A} \rangle) &= (\text{proj}_2, E \times_B A)(\text{proj}_2, E \times_B (E \times_B A))(\langle \text{proj}_1, 1_{E \times_B A} \rangle) = \\
 &= (\text{proj}_2, E \times_B A)(1_{E \times_B A}) = \text{proj}_2, E \times_B A = (\text{proj}_2, E \times_B A)(\text{Id}_{(E \times_B A, \text{proj}_1)})
 \end{aligned}$$

and the second composition

$$(\text{proj}_1, E \times_B A)(\tilde{\text{proj}}_2)(\langle \text{proj}_1, 1_{E \times_B A} \rangle) = (\text{proj}_1, E \times_B (E \times_B A))(\langle \text{proj}_1, 1_{E \times_B A} \rangle) =$$

$$\text{proj}_{1, E \times_B A} = (\text{proj}_{1, E \times_B A})(\text{Id}_{(E \times_B A, \text{proj}_1)})$$

Finally we can check the second triangle identities. Let (D, δ) be an object in $(\mathcal{A} \downarrow E)$, we have:

$$\begin{aligned} ((\epsilon p!) \circ (p! \eta))_{(D, \delta)} &= ((\epsilon p!)_{(D, \delta)}(p! \eta))_{(D, \delta)} = (\epsilon_{p!(D, \delta)} p!)(\langle \delta, 1_D \rangle) = \\ &= (\epsilon_{(D, p\delta)}) p! (\langle \delta, 1_D \rangle) = \text{proj}_{2, E \times_B D}(\langle \delta, 1_D \rangle) = 1_D = \text{Id}_{(D, \delta)} \end{aligned}$$

$$\begin{array}{ccc} E \times_B D & \xrightarrow{\text{proj}_{2, E \times_B D}} & D \\ \downarrow \text{proj}_{1, E \times_B D} & & \downarrow p\delta \\ E & \xrightarrow{p} & B \end{array}$$

Now we can conclude that

$$\begin{array}{ccc} & \xrightarrow{p^*} & \\ (\mathcal{A} \downarrow B) & \perp & (\mathcal{A} \downarrow E) \\ & \xleftarrow{p!} & \end{array}$$

is an adjunction.

Bibliografia

[1] bla bla bla